(*) Recursively defined sequences.

Example:

$$a_1 = 1$$

$$a_{n+1} = 1 + \frac{1}{a_n}$$

Find $\lim_{n\to\infty} a_n$ assuming the limit exists.

Suppose $\lim_{n\to\infty} a_n = L$

$$\Rightarrow \lim a_{n+1} = \lim\left(1 + \frac{1}{a_n}\right)$$

$$\Rightarrow \lim a_{n+1} = \lim 1 + \lim\left(\frac{1}{a_n}\right)$$

$$\lim a_{n+1} = 1 + \frac{1}{\lim a_n}$$

$$\Rightarrow L = 1 + \frac{1}{L}$$

multiply both sides by L

$$L^2 = L + 1$$

$$L^2 - L - 1 = 0$$

$$b^2 - 4ac = 1 + 4 = 5$$

$$\Rightarrow \quad L = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{1 \pm \sqrt{5}}{2}$$

$$L = \nearrow \frac{1+\sqrt{5}}{2}$$

$$\searrow \frac{1-\sqrt{5}}{2}$$

$$a_1 = 1$$

$$a_{n+1} = 1 + \frac{1}{a_n}$$

Because $a_n > 0 \Rightarrow \lim a_n = L = \frac{1+\sqrt{5}}{2}$

Example: $a_1 = 1$

$$a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right)$$

Suppose $\lim a_n$ exists. Find the limit.

Suppose: $\lim a_n = L$

$$\Rightarrow \quad \lim a_{n+1} = \lim \frac{1}{2}\left(a_n + \frac{2}{a_n}\right)$$

$$\Rightarrow \quad \lim a_{n+1} = \frac{1}{2}\lim a_n + \frac{1}{2}\cdot\frac{2}{\lim a_n}$$

$$\Rightarrow \quad L = \frac{1}{2}\cdot L + \frac{1}{L}$$

$$\Rightarrow \quad L - \frac{1}{2}L = \frac{1}{L}$$

$$\Rightarrow \quad \frac{1}{2}L = \frac{1}{L} \quad \Rightarrow L^2 = 2$$

$$\Rightarrow \quad L = \pm\sqrt{2}$$

B/C $a_n > 0 \Rightarrow L = \sqrt{2} \Rightarrow \lim a_n = \sqrt{2}$